

Physics-Informed Identification of the MF6.2 Pure-Lateral Tyre Model

from Skidpad Telemetry and Calspan TTC Data

Algorithm, vehicle physics, data audit
and an identifiability analysis of the P26 PINN

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Contents

Abstract **3**

1 Introduction **5**

2 Physics **6**

2.1 Sign conventions 6
2.2 MF6.2 pure lateral force 7
2.3 Slip kinematics 7
2.4 Load transfer 8
2.5 Steady-state balances: the telemetry as an axle-force sensor 8

3 The Algorithm **9**

3.1 Parameters 9
3.2 Composite loss 9
3.3 Optimisation schedule 10

4 Data Audit **11**

4.1 TTC data: reference cornering sweeps and a provenance hazard 11
4.2 Telemetry 12

5 Reference Results **14**

5.1 Convergence and loss balance 14
5.2 TTC fit 14
5.3 Identified tyre 15
5.4 Vehicle level 17
5.5 A deep-learning engineer's reading of the reference run 20
5.6 Re-running on the archived data 21

6 Diagnosis **21**

6.1 The saturation manifold 21
6.2 Errors-in-variables attenuation: a controlled experiment 22
6.3 What the TTC data can and cannot identify 23
6.4 Hessian spectrum and gradient conflict 24
6.5 The steer-induced yaw moment has the wrong sign 25

7 Controlled Experiments **25**

8 Roadmap **27**

8.1 What the present data support 27
8.2 Data 27

8.3 Model and algorithm 28

9 Conclusion **28**

Acknowledgments **29**

A Identified Coefficients **30**

B Output of the Default Re-run (Run A) **30**

C Experiment Harness **31**

References **32**

Abstract

This paper documents and critically evaluates a physics-informed learning pipeline ("PINN") that estimates the pure-lateral Magic Formula 6.2 (MF6.2) coefficients of the P26 Formula Student tyre (Hoosier 43075). The pipeline fuses two independent data streams: Calspan Tire Test Consortium (TTC) cornering sweeps (±12°, five loads from 217 to 1126 N, 6 596 samples) and 23 238 samples of steady-state skidpad telemetry on 10.5, 9 and 6 m circles. The 17 tyre coefficients, two per-axle grip scalars and a lateral-velocity sensor offset are float64 PyTorch parameters. They are fitted jointly by minimising a composite loss: the TTC force residual, the vehicle lateral-force and yaw-moment balances, and a physics-motivated prior. The loss passes through a differentiable port of the team's MATLAB load-transfer model and a rigid-body slip kinematics model, and is minimised with Adam (cosine schedule) followed by full-batch L-BFGS.

Reference results. The TTC fit is excellent ($R^2 = 0.994$, RMSE 106 N) at every load. The identified tyre peaks at 7–11° with degressive friction ($p_{Dy2} < 0$) and a cornering stiffness of 26.2 kN/rad at $F_{z0} = 665.5$ N. The per-axle grip scalars converge to $\mu_f = 0.561$ and $\mu_r = 0.576$, i.e. on-track grip at about 56–58% of the TTC belt. The vehicle-level fit is much weaker: the lateral-velocity offset settles on its +2.50 m/s bound, the predicted a_y is almost constant ($R^2 = -0.16$) and the operating slip angles lie at −15° to −25°, beyond both the tyre peak and the TTC data range.

Diagnosis. A systematic audit, backed by controlled experiments, explains this gap:

1. **The steady-state balances admit a degenerate "saturated-tyre" solution.** Kinematic slip angles fluctuate by 1.2–3.8° within a run while the axle forces stay essentially constant. Under these errors-in-variables conditions, least squares minimises the loss by shifting every operating point onto the flat, post-peak part of the curve (through the offset) and scaling the plateau to a_y/g (through the grip scalars). A controlled simulation reproduces this attenuation from slip-angle noise alone.

2. **Data and code defects.** The steering channel of the four 6 m runs has the wrong sign. The steer-induced yaw-moment term has the opposite sign to the geometry used everywhere else. The grip scalar multiplies cornering stiffness as well as friction, and hard clamps create zero-gradient attractors. In addition, the TTC data currently archived with the pipeline are a *drive/brake* test with the same file names as the reference cornering sweeps. Re-running on them degrades the TTC fit to $R^2 = 0.81$.

3. **Loss balance.** At convergence the TTC term contributes only 7% of the objective; the prior contributes 38%, 90% of which is the offset's anchor.

Correcting the data and code in controlled re-runs lifts $R^2(a_y)$ from about 0 to 0.32, but does *not* remove the offset (≈ 1.4 m/s). This is the expected outcome while the lateral velocity is measured less accurately than the slip angles it must resolve. We therefore separate what is identified (the TTC shape and a track-to-TTC grip ratio of ≈ 0.56–0.67) from what is not (the on-track operating slip angles, and hence the on-track stiffness and peak slip), and give a concrete data and modelling roadmap.

Keywords: Physics-informed learning, Magic Formula 6.2, tyre identification, errors-in-variables, identifiability, vehicle dynamics, Formula Student

1 Introduction

The lateral force of the tyre is the single most influential constitutive law in a vehicle-dynamics model. It sets the limit lateral acceleration, the understeer gradient, the transient yaw response and, through the controllers that rely on it, the behaviour of torque-vectoring and stability systems. Formula Student teams usually take it from the TTC database. That data is measured on a flat-belt machine with an abrasive surface, and its friction level typically overstates asphalt by a factor close to two.

The P26 pipeline analysed here tries to close that gap. It keeps the TTC data as the primary supervision, but also requires the *same* tyre model, applied to all four wheels of the car, to reproduce the lateral and yaw equilibrium measured on the skidpad. In principle this is an elegant idea. The vehicle equations act as a *physics residual*, just as a partial differential equation does in a classical physics-informed neural network (Raissi et al., 2019), and the telemetry adds information about the track surface that the laboratory cannot provide.

Strictly speaking, the model contains no neural network. It is a *differentiable-physics grey-box* identification problem: every learnable quantity has a physical meaning, and automatic differentiation plus stochastic optimisation replace a hand-written Gauss–Newton solver. The PINN vocabulary is still useful, because the method inherits both the strengths of physics-informed training (constraints, data fusion) and its known pathologies. The most important of these are imbalanced loss terms (Wang et al., 2021) and a physics residual that can be satisfied by a family of degenerate solutions.

What this paper does.

1. Derives the physics used by the code: MF6.2 pure lateral, rigid-body slip kinematics, Milliken load transfer and the steady-state balances (Section 2).

2. Formalises the algorithm: parameterisation, loss, priors and optimisation schedule, with their Bayesian reading (Section 3).

3. Audits the data: the reference TTC cornering sweeps, a data-provenance hazard in the archived rig data, a steering-sign error and slip-angle noise in the telemetry (Section 4).

4. Presents the reference results and reads them from a deep-learning-engineering perspective (Section 5).

5. Diagnoses the failure: saturation manifold, errors-in-variables attenuation, TTC non-identifiability, Hessian spectrum and gradient conflict (Section 6).

6. Tests six controlled variants of the pipeline (Section 7).

7. Gives a data and modelling roadmap (Section 8).

Sources and reproducibility. Two kinds of evidence are used. (i) The *reference results*: the training outputs of the production run (tyre curves, TTC fit, per-wheel time series, vehicle balance, operating points) and the configuration, loss values and coefficient set documented in the team presentation *Physics-Informed Neural Network for Tyre Discovery* (Prom Racing, 2026). (ii) *Controlled re-runs* performed for this paper. The pipeline was used *unmodified*, and every variant in Section 7 is a documented change applied on top of it. The re-runs use the same nine processed skidpad runs as the reference run (23 238 samples). The TTC cornering sweeps of the reference run were not part of the archived data, so the re-runs used the TTC data that were (Section 4).

2 Physics

2.1 Sign conventions

The telemetry uses a body frame with x forward, and in all runs $a_y < 0$ and $r < 0$ with the right-hand wheels carrying more load. That is a left-hand (counter-clockwise) turn in the SAE frame (y right, z down), or equivalently a right-hand turn in ISO. Because the two are mirror images, the choice only matters for *labelling*. What must hold is internal consistency: every block of the model must place the wheel labelled "FL" on the same side of the car. Section 6.5 shows that one block does not.

The tyre model uses the convention "positive slip angle $\Rightarrow$ positive F_y". The TTC data follow SAE (positive SA gives negative FY), so the loader negates FY and keeps SA. For a tyre curve that is an odd function this is exact. For the asymmetric terms (S_{Hy}, S_{Vy}, p_{Ey3}) it is not, which matters when the same curve is applied to the left and right wheels of the car.

2.2 MF6.2 pure lateral force

With camber γ, normalised load $\mathrm{d}f_z = (F_z - F_{z0})/F_{z0}$ and all scaling factors equal to one, the code implements (Pacejka, 2012; Besselink et al., 2010)

$$F_{y0} = D_y \sin\big[C_y \arctan\{B_y\alpha_y - E_y(B_y\alpha_y - \arctan B_y\alpha_y)\}\big] + S_{Vy}, \tag{1a}$$

$$\alpha_y = \alpha + S_{Hy}, \qquad S_{Hy} = p_{Hy1} + p_{Hy2}\mathrm{d}f_z + p_{Hy3}\gamma, \tag{1b}$$

$$C_y = p_{Cy1}, \qquad D_y = \mu_y F_z, \qquad \mu_y = (p_{Dy1} + p_{Dy2}\mathrm{d}f_z)(1 - p_{Dy3}\gamma^2), \tag{1c}$$

$$K_{y\alpha} = p_{Ky1}F_{z0}\sin\Big(2\arctan\frac{F_z}{p_{Ky2}F_{z0}}\Big), \qquad B_y = \frac{K_{y\alpha}}{C_y D_y}, \tag{1d}$$

$$E_y = (p_{Ey1} + p_{Ey2}\mathrm{d}f_z)\,\{1 - (p_{Ey3} + p_{Ey4}\gamma)\,\mathrm{sgn}(\alpha_y)\} \quad (\le 1), \tag{1e}$$

$$S_{Vy} = F_z\{p_{Vy1} + p_{Vy2}\mathrm{d}f_z + (p_{Vy3} + p_{Vy4}\mathrm{d}f_z)\gamma\}. \tag{1f}$$

The geometric meaning of B, C, D, E is the same as for the longitudinal model: the slope at the origin is $B_yC_yD_y = K_{y\alpha}$; the peak D_y is reached at $\varphi^\star = \tan(\pi/2C_y)$ provided $C_y > 1$; the large-slip asymptote is $D_y\sin(C_y\pi/2)$; and $E_y \le 1$ is required for a monotone inner function.

The cornering-stiffness law. Equation (1d) is a saturating load law. $\sin(2\arctan u) = 2u/(1+u^2)$ reaches its maximum at $u = 1$, so $K_{y\alpha}$ peaks at $F_z = p_{Ky2}F_{z0}$ with value $p_{Ky1}F_{z0}$, and becomes *degressive* for higher loads. With $F_{z0} \approx 724$ N and $p_{Ky2} \approx 2.8$–3.2, the peak sits near 2–2.3 kN, far above the operating range. Over the data, $K_{y\alpha}$ is therefore almost linear in F_z, and p_{Ky1} and p_{Ky2} are strongly correlated.

Camber terms. All TTC samples have $\gamma \equiv 0$, and the vehicle model never passes a camber angle. The gradients of p_{Dy3}, p_{Ey4}, p_{Hy3}, p_{Vy3} and p_{Vy4} are therefore identically zero. These five parameters are *structurally non-identifiable*, and their reported values are simply their initial values (or whatever the prior imposes, for p_{Ey4}).

2.3 Slip kinematics

For a rigid body moving with CoG velocity (V_x, V_y) and yaw rate r, the velocity of a contact point at (x_i, y_i) is $(V_x - ry_i,\ V_y + rx_i)$. In the frame the code uses, FL is at $(l_f, -T_f/2)$, so

$$\alpha_{FL} = \delta - \arctan\frac{V_y + l_f r}{V_x + \tfrac{1}{2}T_f r}, \qquad \alpha_{RL} = -\arctan\frac{V_y - l_r r}{V_x + \tfrac{1}{2}T_r r}, \tag{2}$$

and analogously for the right wheels, with $-\frac{1}{2}Tr$ in the denominator. The model neglects steering compliance, Ackermann geometry, static toe and the lever arm between the velocity sensor and the CoG. The last of these matters: a sensor x_s ahead of the CoG reads

$V_{y,s} = V_y + x_s r$. On a skidpad, where r is nearly constant, that error is indistinguishable from a constant offset $v_{y,\text{off}} = -x_s r$.

2.4 Load transfer

Wheel loads follow the team's MATLAB load-transfer routine: static loads, then longitudinal transfer split into sprung and unsprung parts, then Milliken's lateral transfer (Milliken and Milliken, 1995), then aerodynamic downforce with a roll-dependent $C_L A$. For the front axle,

$$\Delta F_{z,f} = a_y \left[\frac{m_s}{T_f} \left(\frac{q\,\kappa_f}{K_\phi - m_s q} + \frac{b}{L} h_{rc,f} \right) + \frac{m_{u,f}}{T_f} h_u \right], \qquad \kappa_f = K_{\phi f} - \frac{b}{L} m_s q, \tag{3}$$

with $q = h_s - h_{rc}$ the roll-moment arm and K_ϕ the total roll stiffness derived from the roll gradient.

Two checks are worth recording. (i) Summing front and rear moments gives $\Delta F_{z,f} T_f + \Delta F_{z,r} T_r = a_y (m_s h_s + m_u h_u)$ exactly, so the *total* transfer is correct whatever the stiffness split. (ii) The code computes K_ϕ from the mass m_s instead of the weight $m_s g$. Because the formula only uses the ratio $\kappa_f / (K_\phi - m_s q)$, this cancels except in the small $m_s q$ correction. The front share of elastic transfer comes out 0.521 with the code and 0.520 with consistent units, a negligible difference.

2.5 Steady-state balances: the telemetry as an axle-force sensor

The two physics residuals are

$$R_{\text{lat}} = m a_y - \left[(F_{y,FL} + F_{y,FR}) \cos\delta + F_{y,RL} + F_{y,RR} \right], \tag{4}$$

$$R_{\text{yaw}} = I_z \dot{r} - \left[l_f (F_{y,FL} + F_{y,FR}) \cos\delta - l_r (F_{y,RL} + F_{y,RR}) + M_\delta \right], \tag{5}$$

where M_δ is the moment produced by the difference between the left and right front forces through the steer angle (Section 6.5). Solving the two balances for the axle forces gives

$$F_{y,f} \approx \frac{m a_y l_r + I_z \dot{r}}{L}, \qquad F_{y,r} \approx \frac{m a_y l_f - I_z \dot{r}}{L}. \tag{6}$$

Key observation

Near steady state, the telemetry *measures the two axle forces* directly from a_y and $\dot{r}$, independently of any tyre model. The tyre model only enters through the slip angles (2) and wheel loads at which those forces must be produced. The information the telemetry adds is therefore a set of points $(\alpha_{\text{axle}},\ F_{y,\text{axle}}/F_{z,\text{axle}})$, an *axle characteristic*, and its value is limited by how accurately α is measured.

A direct consequence: the static weight distribution satisfies $F_{z,f}/F_{z,r} = l_r/l_f$, so if both axles operate at the *same* friction utilisation $\mu_f = \mu_r = a_y/g$, the yaw balance holds automatically. That is exactly what a saturated tyre (every wheel on its plateau) produces. Section 6.1 shows that the optimiser finds this solution.

3 The Algorithm

3.1 Parameters

The learnable vector is

$$\theta = \big(\underbrace{p_{Cy1}, p_{Dy1..3}, p_{Ey1..4}, p_{Ky1,2}, p_{Hy1..3}, p_{Vy1..4}}_{17\ \text{MF6.2}},\ \lambda_f,\ \lambda_r,\ v_{y,\text{off}}\big) \in \mathbb{R}^{20}, \tag{7}$$

where $\lambda_{f,r} = \text{softplus}(\cdot) + 0.5 \geq 0.5$ multiply the *whole* tyre force on each axle, and $v_{y,\text{off}}$ is added to the measured lateral velocity. F_{z0} is fixed to the average of the TTC load levels, $I_z = 100\ \text{kg m}^2$ is fixed, and the steering ratio is fixed at 3.28. The forward pass also contains four *hard* operations: $C_y \in [1.40, 1.60]$ (clamp), $E_y \in [-1, 0.99]$ (clamp), $F_z \geq 30$ N (clamp) and $\text{sgn}(\cdot) \approx \tanh(30\,\cdot)$.

3.2 Composite loss

$$\mathcal{L}(\theta) = w_{\text{ttc}}\mathcal{L}_{\text{ttc}} + w_{\text{lat}}\mathcal{L}_{\text{lat}} + w_{\text{yaw}}\mathcal{L}_{\text{yaw}} + w_{\text{reg}}\mathcal{R}, \qquad (w_{\text{ttc}}, w_{\text{lat}}, w_{\text{yaw}}, w_{\text{reg}}) = (3, 8, 6, 10^{-3}). \tag{8}$$

- **TTC term.** $\mathcal{L}_{\text{ttc}} = 0.7\,\frac{1}{S}\sum_s \frac{\overline{(F_y^{\text{MF}} - F_y)^2}_s}{\bar{F}_{z,s}^2} + 0.3\,\frac{\overline{(F_y^{\text{MF}} - F_y)^2}}{(1000\,\text{N})^2}$. The first part weights every load file equally in friction-coefficient space, the second in force space.

- **Balance terms.** $\mathcal{L}_{\text{lat}} = \sum_i w_i\ \text{Huber}_{0.15}(R_{\text{lat},i}/mg)/\sum_i w_i$ and $\mathcal{L}_{\text{yaw}}$ likewise with $\text{Huber}_{0.25}(R_{\text{yaw}}/1000\,\text{N m})$. The weights $w_i = \exp[-(\dot{r}_i/1\,\text{rad s}^{-2})^2]$ are a soft steady-state filter.

- **Prior $\mathcal{R}$.** Quadratic pulls toward zero (e.g. $150\,p_{Ey3}^2$, $30\,v_{y,\text{off}}^2$, $50\,p_{Vy1}^2$), toward one ($10(\lambda-1)^2$), and one-sided quadratic walls (e.g. $300\,\text{relu}(1.4-p_{Cy1})^2$, $500\,\text{relu}(0.7-\lambda)^2$, $5\,\text{relu}(55-p_{Ky1})^2$).

Bayesian reading. Up to constants, $\mathcal{L}$ is a negative log-posterior. Each quadratic term $c_j(\theta_j - \bar{\theta}_j)^2$ is a Gaussian prior whose width shrinks as $w_{\text{reg}}c_j$ grows, and each one-sided wall is a soft truncation of that prior. The strength of a prior is only meaningful *relative to the curvature of the data terms*. With $w_{\text{reg}} = 10^{-3}$, a wall of weight 500 costs 5×10^{-3} for a 0.1 violation, the same order as the balance losses it is supposed to police ($\approx 10^{-2}$).

This is why several "hard" boxes are violated in practice (Section 5).

3.3 Optimisation schedule

Algorithm 1 P26 PINN training (reference configuration)

Input: TTC tensors, telemetry tensors, θ_0 (Table 5, "init"); all tensors `float64`

1: **Phase 0**: Adam on tyre parameters only, loss $= \overline{(F_y^{\mathrm{MF}} - F_y)^2}/10^6 + w_{\mathrm{reg}}\mathcal{R}_{\mathrm{tyre}}$ (TTC-only warm-up)

2: **Phase 1** (5000 epochs): Adam (Kingma and Ba, 2015), $\eta_0 = 3{\cdot}10^{-3}$, cosine annealing to $\eta_{\min} = 10^{-5}$; telemetry mini-batch 512, TTC full batch; gradient-norm clip 5

3: every 50 epochs: record the *mini-batch* loss; store the state if it is the lowest so far

4: **Phase 2**: reload the best stored state; L-BFGS (Nocedal and Wright, 2006) on the full batch, history 20, strong-Wolfe line search, until the convergence tolerance

5: every 5 closure calls: record the *full-batch* loss; store the state if it is the lowest so far

6: reload the best stored state ▷ mini-batch and full-batch losses are compared

Output: $\theta^\star$, plots, run summary

Table 1: Reference training configuration (team presentation) versus the defaults hard-coded in the current code, which the controlled re-runs in Section 7 use.

Setting	Reference run	Code defaults
Adam epochs / warm-up	5000 / TTC-only	5000 / 1200
Learning rate	$3 \cdot 10^{-3} \to 10^{-5}$ (cosine)	$3 \cdot 10^{-3} \to 1.5 \cdot 10^{-4}$ (cosine, $0.05\eta_0$)
Batch (telemetry / TTC)	512 / full	2048 / 1024
L-BFGS	history 20, to tolerance	history 50, ≤ 800 iterations
Loss weights $w_{\mathrm{ttc}}, w_{\mathrm{lat}}, w_{\mathrm{yaw}}, w_{\mathrm{reg}}$	3, 8, 6, 10^{-3}	3, 8, 6, 10^{-3}
F_{z0}	665.5 N (mean of 5 TTC loads)	mean of the TTC files supplied
Precision	`float64`	`float64`

`float64` is a deliberate and sound choice. Near $\alpha = 0$ the curvature term $B_y\alpha_y - \arctan(B_y\alpha_y) \approx (B_y\alpha_y)^3/3$ suffers catastrophic cancellation in single precision, which corrupts the gradients of E_y exactly where the stiffness is identified.

The final line of Algorithm 1 is a latent defect. A lucky mini-batch loss from Phase 1 can be lower than any full-batch loss from Phase 2, and the L-BFGS result is then discarded. In our default-configuration re-run the reported "best loss" (0.6725) is indeed a mini-batch value, whereas L-BFGS reaches 0.69457 on the full objective. With the smaller reference batch (512) the mini-batch losses are noisier, so the risk is higher. The training-history plot of the reference run shows the L-BFGS spike at epoch ≈ 5000, but not whether that state was kept.

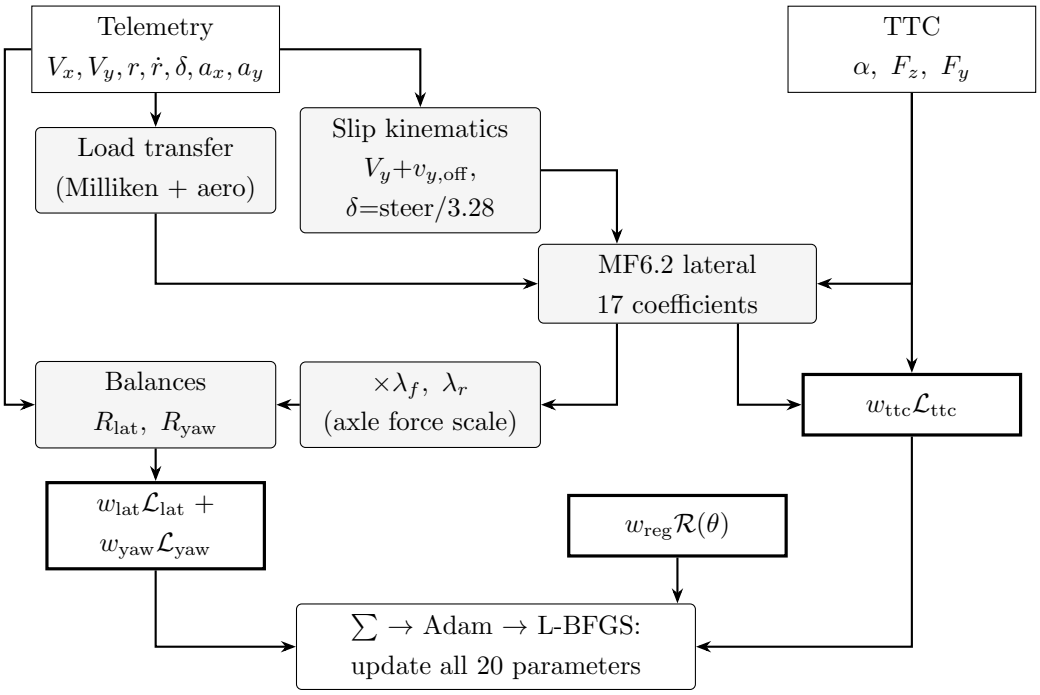


Figure 1: Computational graph of the P26 PINN. The same MF6.2 instance serves the TTC term (unscaled) and all four wheels (scaled by $\lambda_{f,r}$). Slip angles depend on the learnable offset $v_{y,\text{off}}$.

4 Data Audit

4.1 TTC data: reference cornering sweeps and a provenance hazard

Reference data. The reference run uses five Calspan cornering sweeps (Round 5) of the Hoosier 43075 FSAE slick at 25 mph, 10 psi and zero inclination, at nominal loads of 50, 100, 150, 200 and 250 lbs, i.e. $F_z \approx 217$, 438, 661, 885 and 1126 N. The slip angle is swept continuously over $\pm 12°$ in free rolling, with the SAE signs converted to ISO ($F_{y,\text{ISO}} = -F_{y,\text{SAE}}$). After rejecting samples with $F_z < 100$ N, 6 596 rows remain. This is a well-designed data set for a pure-lateral MF. Both slip signs are covered, so the asymmetry terms are identifiable in principle. The peak (at 7–11°) and part of the post-peak range lie inside the data, and five loads span the on-car range except its extremes ($F_z^{\min} = 77$ N on the unloaded front wheel, $F_z^{\max} = 1351$ N on the loaded rear).

Provenance hazard. The four TTC data sets currently archived with the pipeline carry the *same run designation* (Round 5, 25 mph, 10 psi, 0°, at 50, 150, 200 and 250 lbs) but contain a different test: a drive/brake run (Figure 2).

- The slip angle takes only **three values**, 0°, −3° and −6°, all on one side.

- At each slip angle the **slip ratio is swept** from -0.21 to $+0.14$, and the median $|F_X/F_Z|$ is 1.4–2.0.
- Only **2.8%** of the 7 108 samples (202 rows) have $|F_X/F_Z| < 0.2$, i.e. are close to pure lateral.

Along a slip-ratio sweep at fixed slip angle, F_Y falls by up to a factor of three (friction-ellipse coupling). A pure-lateral MF fitted to these data learns the *conditional mean over slip ratio*, and the peak slip angle lies outside the data. Anyone who re-runs the pipeline on the archived data as they stand gets a markedly worse tyre (Section 5). The fix is purely organisational: keep each TTC test type clearly separated, and add a guard to the TTC loader that rejects data whose slip ratio or $|F_X|$ is not near zero.

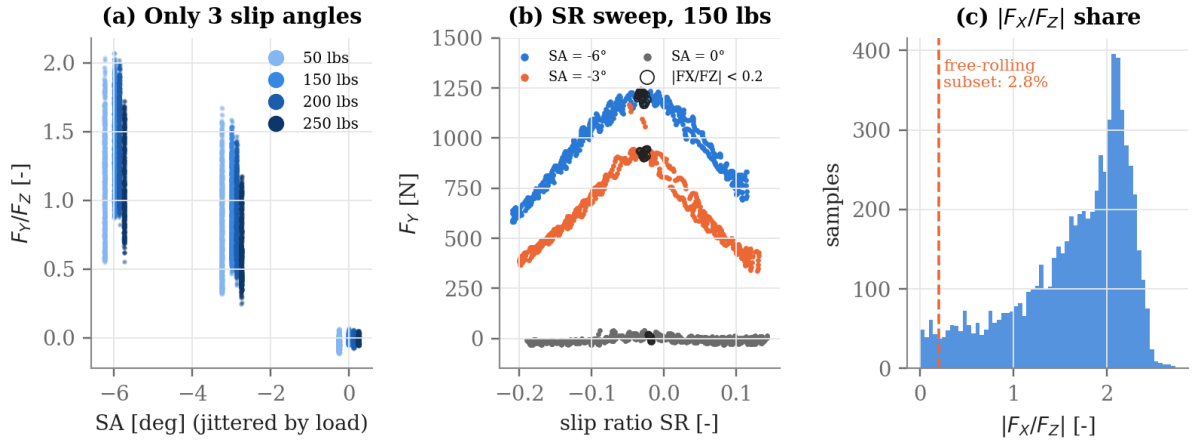


Figure 2: Audit of the TTC data currently archived with the pipeline (not the reference sweeps). (a) Normalised lateral force at the only three slip angles (loads jittered horizontally). (b) Lateral force along the slip-ratio sweep at $-6°$ and $-3°$ (150 lbs); circles mark the near-free-rolling samples. (c) Distribution of $|F_X/F_Z|$: only 2.8% of samples are below 0.2.

4.2 Telemetry

Nine processed runs are used: four on the 10.5 m circle, one on the 9 m circle and four on the 6 m circle. After the speed window (6.7–15.7 m/s) and the steady-state weight, the effective sample is dominated by quasi-steady cornering at 0.75–1.28 g. Three findings stand out (Figure 3).

(1) Steering sign on the 6 m runs. All nine runs turn the same way ($a_y < 0$, $r < 0$), yet the steering channel is *negative* on the 10.5/9 m runs and *positive* on all four 6 m runs. On the tightest circle, which needs the *largest* steer angle, the road-wheel angle has the wrong sign (Figure 3a). The implied understeer gradients are -13 to $-18°/g$, physically impossible, compared with -0.9 to $-2.9°/g$ on the other runs. With the supplied code, the 6 m runs produce front slip angles of $+7°$ to $+11°$ *opposite* to the required force, which

is visible as the spurious yaw-acceleration spikes in the reference balance plot (Figure 9). The in-code comment that the ratio 3.28 was "confirmed by K_{us} regression across all runs" should be read with care. On the 10.5/9 m runs, the Ackermann angle Lr/V_x and $a_y = V_x r$ are nearly collinear (all runs share one radius), so the steering ratio and K_{us} cannot be separated from steady-state data.

(2) The kinematic slip angles are noisy. Within a single run at essentially constant state, the kinematic rear slip angle has a standard deviation of 1.2–3.8°. The kinematic identity $a_y = \dot{V}_y + V_x r$ is violated with a residual standard deviation of 2.9 m/s^2 (Figure 3c), and integrating it over a run lets V_y drift by several m/s. The run-mean V_y also differs between nominally identical runs (0.24 m/s for runs 1–2 versus 0.72 m/s for runs 3–4 on the 10.5 m circle). The V_y channel therefore carries both high-frequency noise and run-to-run bias. Its error is of the same order as the slip angles to be measured.

(3) The axle characteristic is flat. Plotting the model-free rear-axle utilisation (6) against the kinematic rear slip angle (Figure 3b) gives a *horizontal band*. $F_{y,r}/F_{z,r} \approx 1.0$ over $-12° < \alpha_r < -2°$, with correlation 0.49 (front: 0.23). Either the axles operate on their plateau, or, much more likely at this noise level, the slip-angle errors hide the underlying slope. Either way, the telemetry carries almost no information about cornering stiffness or peak slip. What it does determine well is the *friction level*: $\mu_{\text{axle}} \approx a_y/g \approx 0.92$–$1.18$.

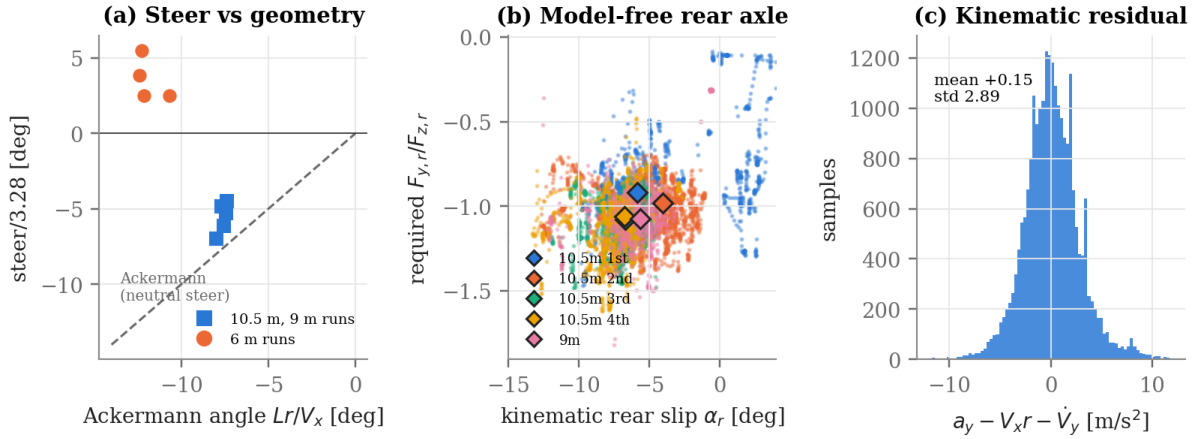


Figure 3: Telemetry audit. (a) Run-mean road-wheel steer (steering/3.28) versus the Ackermann angle: the 6 m runs (circles) have the wrong sign. (b) Model-free rear-axle utilisation versus kinematic slip angle (10.5/9 m runs, $|\dot{r}| < 1$, $|a_x| < 2$; diamonds are run medians). (c) Residual of the kinematic identity $a_y = \dot{V}_y + V_x r$.

(4) Combined slip on track. The P26 is rear-wheel-driven, and the skidpad telemetry has a_x with a standard deviation of 1.6–2.5 m/s^2 (peaks ±7 m/s^2). A pure-lateral tyre

model is only valid where $|a_x|$ is small. Keeping $|a_x| < 2$ m/s^2 retains 75% of the non-6 m samples.

5 Reference Results

5.1 Convergence and loss balance

Table 2 lists the loss components of the reference run at convergence. The weighted total checks out exactly: $3(0.01362) + 8(0.02645) + 6(0.01495) + 10^{-3}(209.95) = 0.5521$.

Table 2: Loss components at convergence (reference run).

Term	Raw value	Weight	Weighted	Share of total
$\mathcal{L}_{\text{ttc}}$ (rig fit)	0.01362	3	0.0409	7.4%
$\mathcal{L}_{\text{lat}}$ (lateral balance)	0.02645	8	0.2116	38.3%
$\mathcal{L}_{\text{yaw}}$ (yaw balance)	0.01495	6	0.0897	16.2%
$\mathcal{R}$ (priors)	209.95	10^{-3}	0.2100	38.0%
$\mathcal{L}$			0.5521	100%

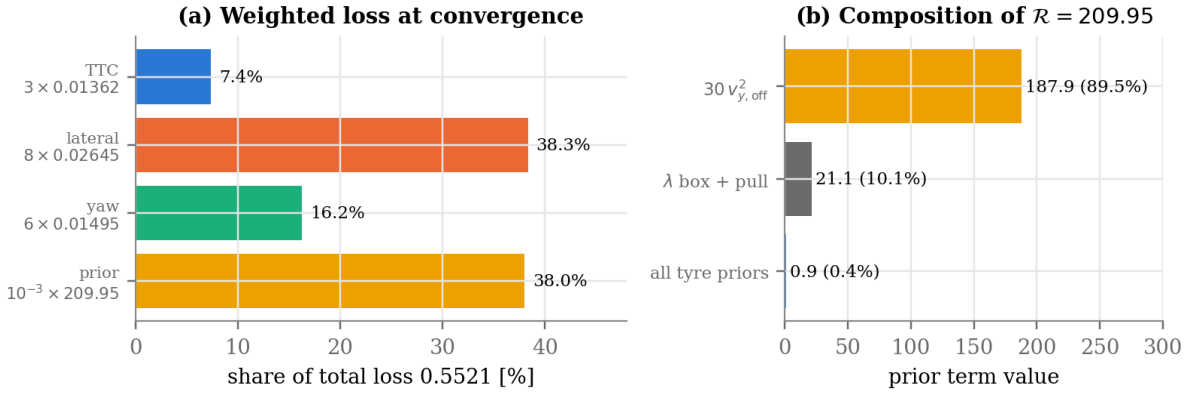


Figure 4: (a) Weighted contribution of each term to the converged objective. (b) Decomposition of the prior $\mathcal{R}$ recomputed from the converged parameters: the offset anchor $30\,v_{y,\text{off}}^2$ alone is 187.9, and the grip-scalar box and pull (both scalars below their soft floor of 0.7) add 21.1. The 15 tyre priors together contribute 0.9.

5.2 TTC fit

The rig fit is excellent at all five loads: RMSE 106.1 N and $R^2 = 0.994$ (Figure 5). The shape, the plateau level and the load dependence are captured. The largest deviations occur on the negative-slip plateau at 150–250 lbs, where the model sits up to about 13%

below the data, and at the highest load near $-4°$. Both are consistent with the asymmetry terms being held near zero by the priors (p_{Ey3} anchor, small S_{Hy}, S_{Vy}).

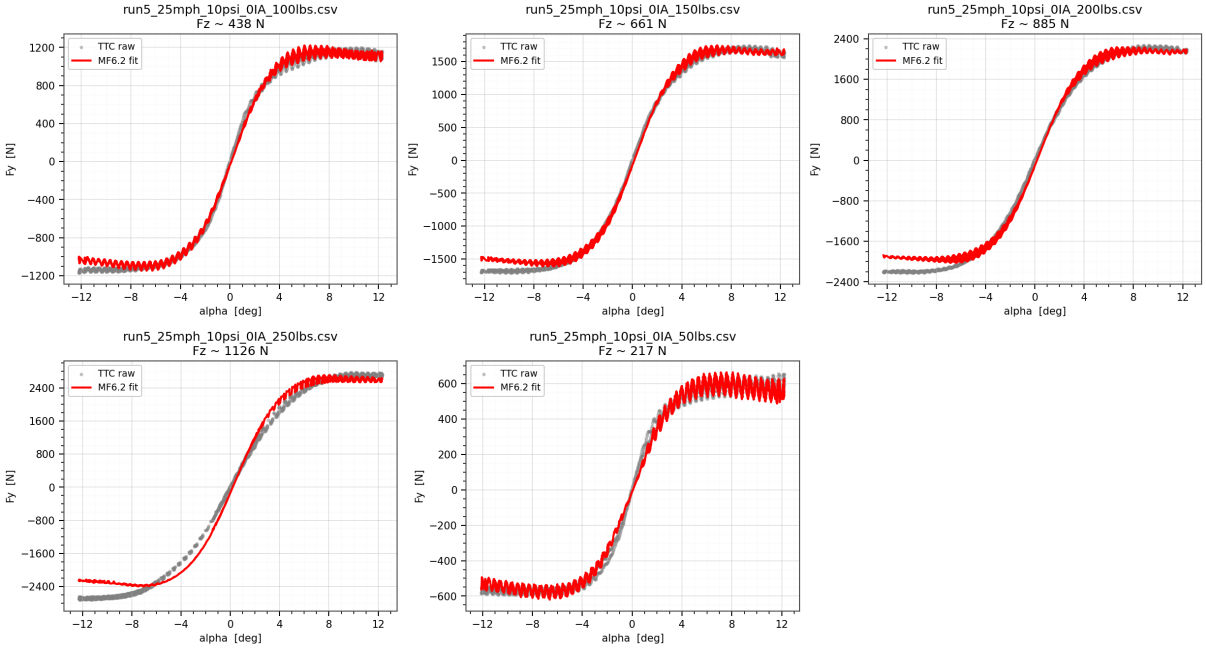


Figure 5: Reference run: MF6.2 fit (red) to the five TTC cornering sweeps (grey), $F_z \approx 217$–1126 N.

5.3 Identified tyre

Table 3 gives the converged coefficient set, and Figure 6 the discovered curves. Evaluating the tabulated set gives the characteristic values: cornering stiffness $K_{y\alpha} = 26.2$ kN/rad (458 N/deg) at F_{z0}, $B_y \approx 11.5$, $C_y = 1.60$, peak $\mu_y = 2.19$ at F_{z0}, falling from 2.35 at 200 N to 2.02 at 1200 N, and peak slip angles from 7.2° to 11.4°. (The presentation quotes "≈ 30 kN/rad". 30 kN/rad is the value at $F_z \approx 800$ N; at F_{z0} the model gives 26.2 kN/rad.)

Table 3: Reference MF6.2 coefficient set ($F_{z0} = 665.5$ N, $I_z = 100$ kg m^2 fixed).

Param.	Value	Note	Param.	Value	Note
p_{Cy1}	1.600	upper clamp	p_{Ky1}	65.86	
p_{Dy1}	2.150		p_{Ky2}	3.012	just above 3.0 box
p_{Dy2}	−0.270	degressive	p_{Hy1}	−0.0037	
p_{Dy3}	0	no data ($\gamma = 0$)	p_{Hy2}	−0.0028	
p_{Ey1}	0.191		p_{Hy3}	0	no data
p_{Ey2}	0.467	near +0.5 box	p_{Vy1}	0.043	
p_{Ey3}	−0.012	anchored	p_{Vy2}	0.050	
p_{Ey4}	0	no data	p_{Vy3}, p_{Vy4}	0	no data
μ_f	0.561	below 0.7 box	$v_{y,\text{off}}$	+2.503 m/s	at +2.5 bound
μ_r	0.576	below 0.7 box	F_{z0}	665.5 N	fixed

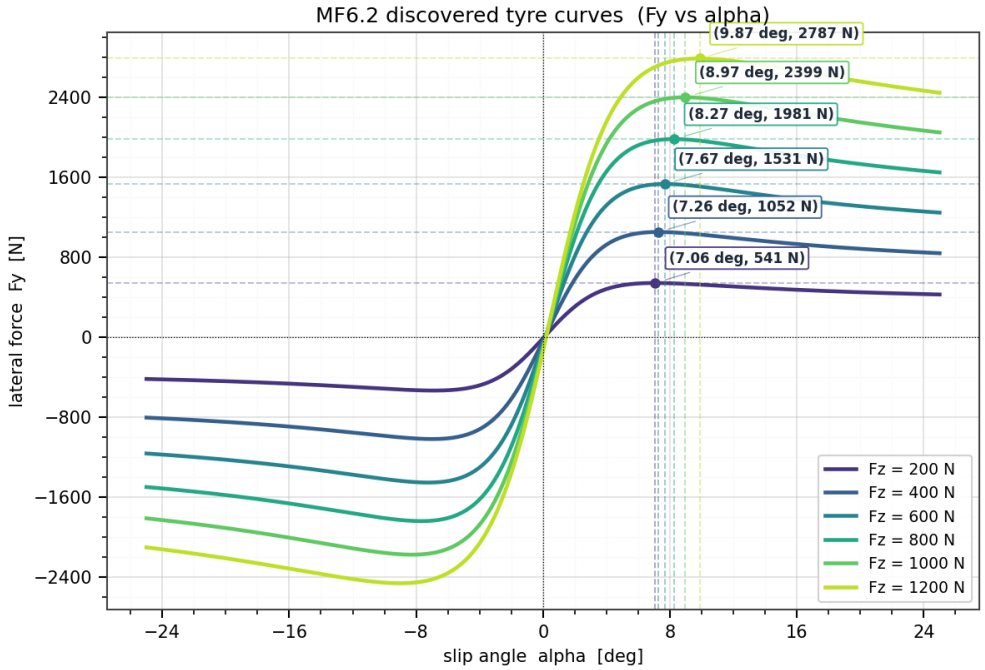


Figure 6: Reference run: discovered MF6.2 curves $F_y(\alpha)$ at 200–1200 N with the positive-side peaks annotated.

Two reference states

The plotted reference outputs (Figures 5–10) and the tabulated coefficient set (Table 3) come from two trainings of the same configuration. They agree in shape but not in level (Figure 7). The plotted run reaches 2787 N at 1200 N (peak $\mu_y = 2.32$, close to the TTC plateau), whereas the tabulated set reaches 2419 N ($\mu_y = 2.02$). Its peak slip angles are also 0.2–1.6° lower. Both fit the TTC data well, which shows that peak friction and the grip scalar trade off along a weak direction: on the car only the product $\mu_{f,r} \cdot \mu_y$ is constrained ($0.561 \times 2.19 \approx 1.23$ at F_{z0} for the tabulated set). This is a direct, practical instance of the identifiability issues analysed in Section 6. We therefore quote the tabulated set as the documented coefficient set and the plotted run as the best-fitting TTC curve.

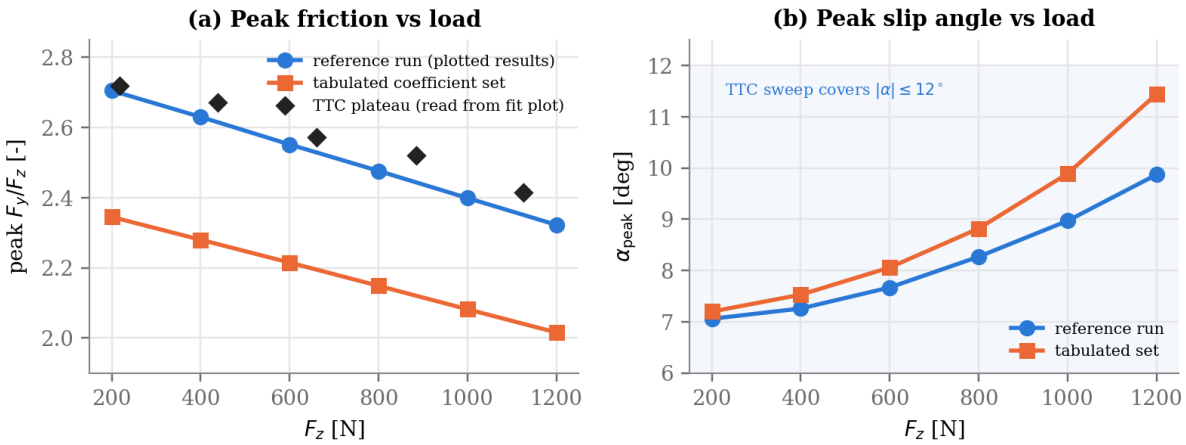


Figure 7: Peak friction and peak slip angle versus load for the plotted reference run and the tabulated coefficient set, with the TTC plateau levels read from Figure 5 (±3%).

5.4 Vehicle level

Figures 8–10 show the vehicle-level outputs. The wheel loads reproduce the MATLAB routine (peaks 747/1218/894/1351 N for FL/FR/RL/RR, minimum 77 N on FL). The picture elsewhere is consistent across all outputs:

- **Slip angles far beyond the peak.** With $v_{y,\text{off}} = +2.50$ m/s, the kinematic slip angles are −12° to −22° on the 10.5/9 m runs; during the 6 m runs the rear goes down to −27° while the front moves towards −2° to −14° (Figure 8). That is two to three times the peak slip angle and beyond the ±12° TTC range.

- **Operating points on the plateau.** The K-means view (Figure 10) shows every cluster on the flat post-peak branch between about −8° and −30°. None lies on the steep part of the curve that the TTC data describe best.

- **Almost constant predicted a_y.** The predicted lateral acceleration sits near -12.5 m/s^2 while the measurement varies between about -8 and -16 m/s^2 (Figure 9). For the tabulated run: $R^2(a_y) = -0.16$ and RMSE 3.20 m/s^2 on all samples; steady-state bias / RMSE -0.68 / 3.39 m/s^2 (lateral) and $+0.26$ / 2.29 rad s^{-2} (yaw). For the plotted run: -1.24 / 3.62 m/s^2 and $+0.16$ / 2.40 rad s^{-2}.

- **6 m runs.** Between 140 and 210 s the front slip angles jump towards zero and the predicted yaw acceleration spikes to 20–25 rad s^{-2}. These are the four 6 m runs whose steering channel has the opposite sign (Section 4).

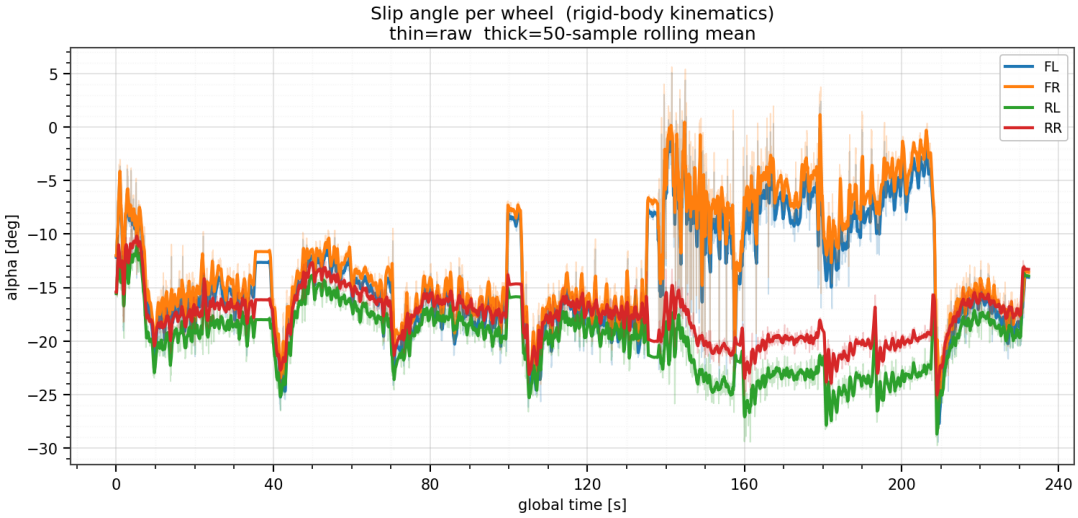


Figure 8: Reference run: kinematic slip angle per wheel (with the learned offset).

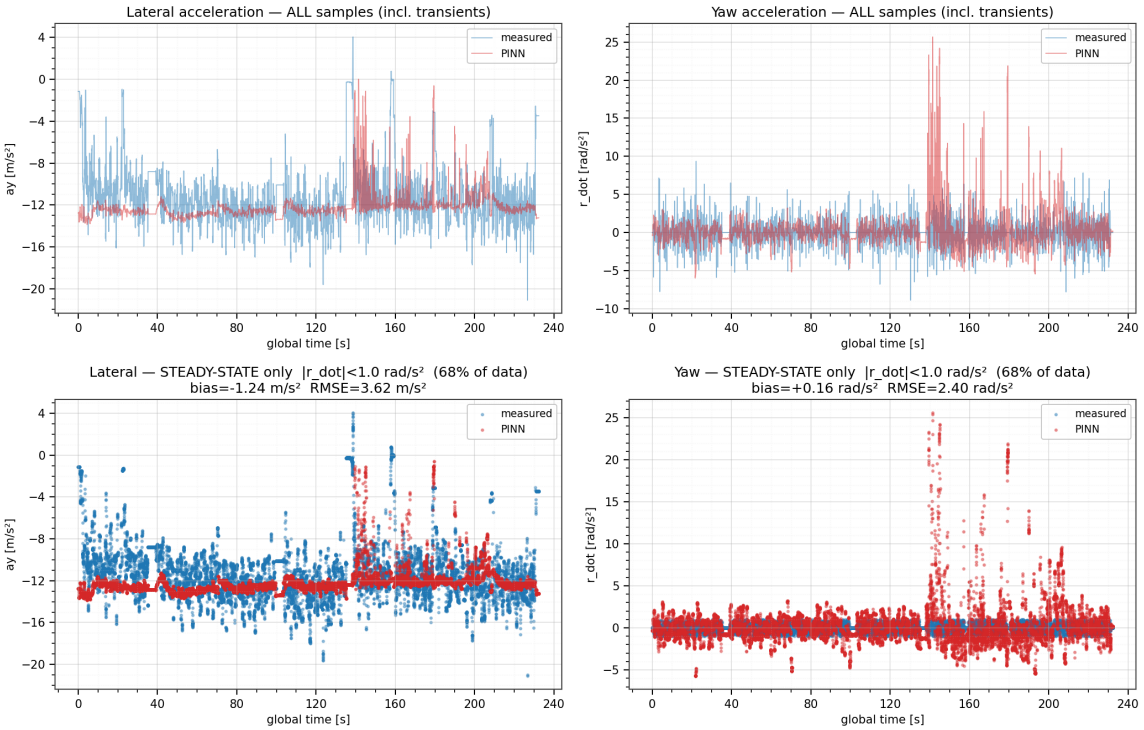


Figure 9: Reference run: measured versus predicted lateral and yaw acceleration, all samples (top) and steady state $|\dot{r}| < 1\ \mathrm{rad\,s^{-2}}$ (bottom).

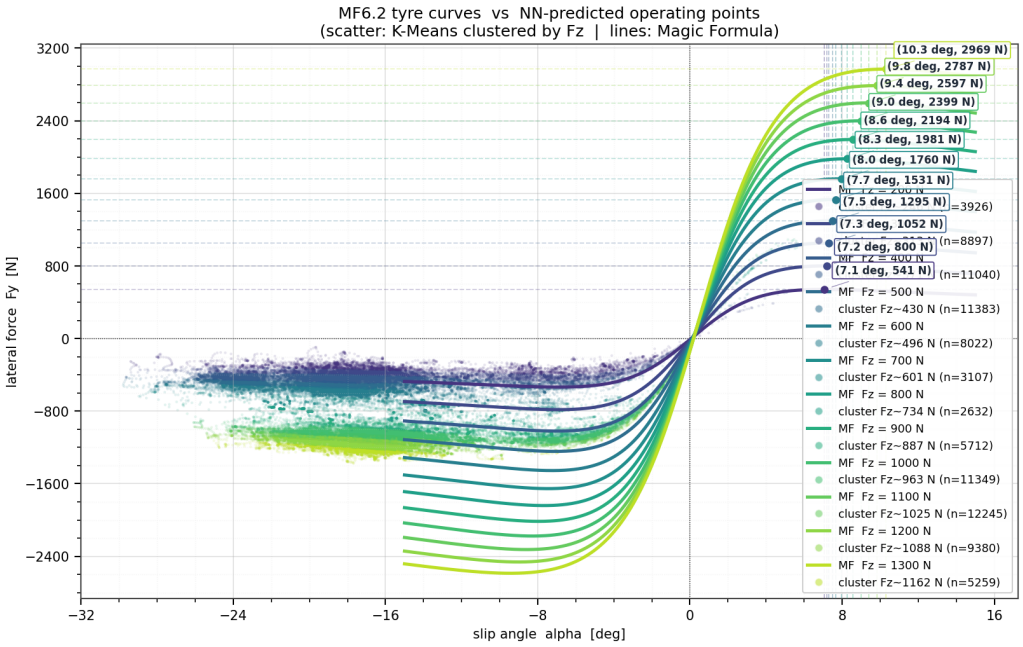


Figure 10: Reference run: MF6.2 curves at 200–1300 N and the on-car operating points, clustered by F_z (K-means).

5.5 A deep-learning engineer's reading of the reference run

1. **The data term is a minority of the objective.** Only 7.4% of the converged loss is the TTC fit. 54.5% is the two balance terms, which the model cannot drive down (it predicts an almost constant a_y), and 38% is the prior, which is itself 90% the offset anchor. Near the optimum, the gradient is therefore dominated by terms that encode a *compromise*, not by information.

2. **A rising prior is a symptom, not a success.** The training history shows $\mathcal{R}$ rising to ≈ 210 and staying there. That is not "physically non-trivial values being found": 187.9 of the 209.95 is $30\, v_{y,\text{off}}^2$ with $v_{y,\text{off}}$ on its wall. A prior that increases monotonically while the data losses plateau signals a parameter being pushed against a constraint.

3. **Soft boxes are not constraints at this weight.** Both grip scalars converge below the $[0.7, 1.3]$ box (0.561, 0.576), where the box costs $\approx 17.4 \times 10^{-3}$. That is less than the balance improvement it buys, so the box is traded away. The same holds for p_{Ky2} (just above 3.0) and p_{Cy1} (on its clamp).

4. **The offset is not a calibrated IMU bias.** The presentation reads $v_{y,\text{off}} = +2.50$ m/s as "the lateral IMU reading high by $\sim 10\%$". At $V_x \approx 11$ m/s this offset is a 13° sideslip error, far beyond any plausible sensor bias or mounting lever arm. Section 6.1 shows the offset is chosen because it moves the operating points onto the plateau.

5. **"56% of TTC grip" is a force scale, not a friction ratio.** Because $\mu_{f,r}$ multiplies the entire force, it also scales the cornering stiffness by 0.56. A friction-only scale gives $\lambda_\mu \approx 0.60$–0.67 in our re-runs (Section 7). The plausible on-track peak friction is $\mu \approx 1.2$–1.35 in both readings.

6. **Batch statistics.** With a telemetry batch of 512 drawn from 23 238 rows that are highly autocorrelated at 100 Hz, one mini-batch covers only a few seconds of *independent* information. The mini-batch loss is noisy (visible in the history), which interacts with the best-state rule of Algorithm 1.

7. **Naming.** There is no neural network in the model: the K-means plot labels "NN-predicted operating points" and the "NN" R^2 metrics (hard-coded to zero) refer to MF6.2 predictions. Renaming avoids confusion when the results are reused.

5.6 Re-running on the archived data

Running the pipeline with its current defaults on the archived data (drive/brake TTC data, all nine telemetry runs) reproduces the same vehicle-level signature, and a weaker tyre. The results are column "A" of Table 4: $v_{y,\text{off}} = +2.502$ m/s (bound), $p_{Cy1} = 1.400$ (lower clamp), $p_{Ey1} = 1.03$ (curvature clamped at 0.99, so the curve has no peak), $p_{Vy1} = 0.225$, TTC $R^2 = 0.81$ and $R^2(a_y) = -0.008$. A second seed returns the same state. The vehicle-level degeneracy therefore appears with both TTC data sets. It is a property of the telemetry and the model, not of the rig data.

6 Diagnosis

6.1 The saturation manifold

Fix the tyre at the run-C solution and scan $v_{y,\text{off}}$ from -0.5 to 3 m/s. At each value, re-optimise λ_f, λ_r to minimise the balance loss alone. Figure 11 shows the result. The balance loss falls *monotonically* by more than an order of magnitude, the standard deviation of the predicted a_y falls from 5.2 to 0.4 m/s^2 (measured: 2.5), and $R^2(a_y)$ rises from -4.6 to $+0.27$, flattening beyond $v_{y,\text{off}} \approx 1.5$ m/s. The same happens when the telemetry is averaged over 2-s windows. The reference run (Section 5) sits at the far end of this valley: $v_{y,\text{off}} = +2.50$ m/s, operating slip angles of $-12°$ to $-27°$, and a predicted a_y that barely moves.

The mechanism follows from Section 2. At $v_{y,\text{off}} = 0$ the operating points lie on the steep part of the curve, where each degree of slip-angle noise becomes several hundred newtons of predicted axle-force noise. A large positive $v_{y,\text{off}}$ moves every wheel onto the plateau, where the predicted force no longer depends on the noisy slip angle. $\lambda_{f,r}$ then sets that plateau to a_y/g, and the yaw balance is satisfied automatically because the static weight distribution already matches l_r/l_f. The optimiser is not malfunctioning. It finds the least-squares answer to a problem whose regressor is noisier than its signal.

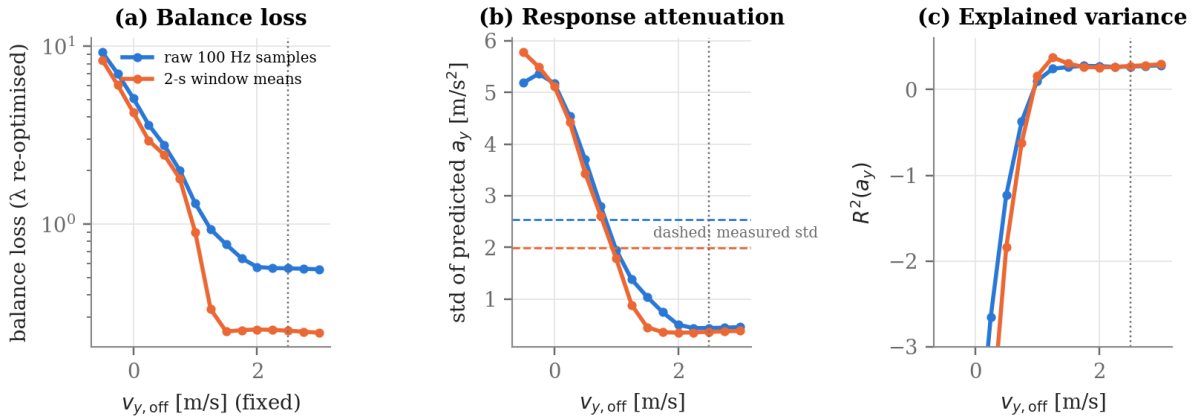


Figure 11: Balance-loss scan over a fixed lateral-velocity offset, with the axle scales re-optimised at each point (tyre from runs E/G). (a) Balance loss. (b) Standard deviation of predicted a_y (dashed: measured). (c) $R^2(a_y)$. The dotted line is the +2.5 m/s wall where run A stops.

6.2 Errors-in-variables attenuation: a controlled experiment

To isolate the mechanism, consider a one-dimensional twin. A known tyre (peak at 6.8°) operates at $\alpha \sim \mathcal{N}(-4.5°, 0.7°)$. The forces are observed almost exactly, but the slip angle is observed with Gaussian noise σ_α. Only a force scale λ and a slip shift s are fitted, with physical bounds. Figure 12 shows that for $\sigma_\alpha = 0$ the fit recovers the truth ($\lambda = 0.999$, $s = 0.00°$). As σ_α grows, the fitted shift pushes the operating points past the peak: to $-6.5°$ at $\sigma_\alpha = 1°$, $-9.1°$ at 2° and $-24.5°$ at 3°. Meanwhile R^2 of the force falls from 0.64 to below zero. The telemetry's measured within-run slip-angle scatter (1.2–3.8°) lies squarely in the range where the twin already fails. This is the classical *regression-dilution* effect (Fuller, 1987; Carroll et al., 2006). Its nonlinear form is especially harmful for saturating functions, because the plateau offers a zero-slope refuge.

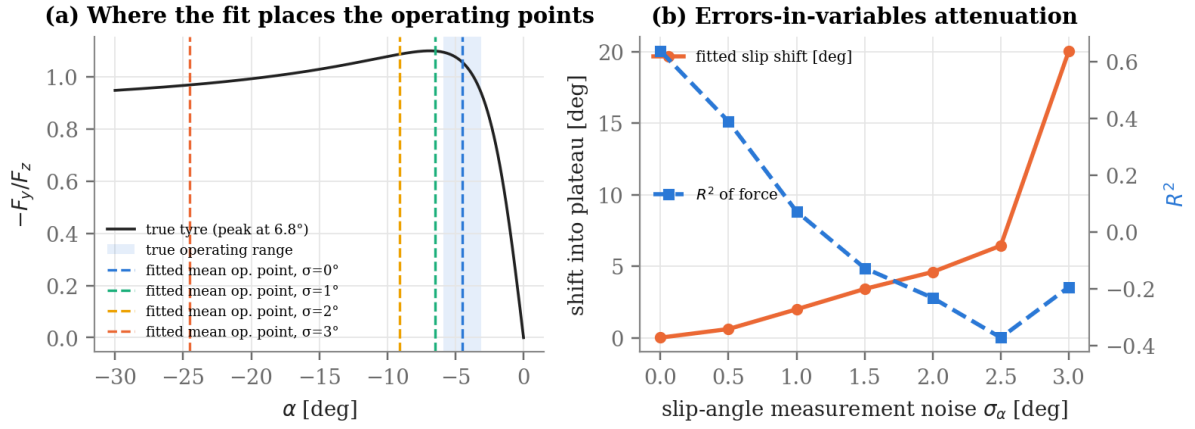


Figure 12: Errors-in-variables twin. (a) True tyre and the mean operating point implied by the fitted shift, for increasing slip-angle noise. (b) Fitted shift and force R^2 versus noise level.

6.3 What the TTC data can and cannot identify

Which TTC file set is used decides how much of the curve the rig can identify. The reference cornering sweeps cover ±12° at five loads, so the peak (7–11°), the plateau, both slip signs and the load dependence are all in the data, and the fit is excellent ($R^2 = 0.994$). Even there, two shape parameters end on constraints ($p_{Cy1} = 1.600$ on the upper clamp, $p_{Ey2} = 0.467$ at the edge of its box), and the peak level trades off against the grip scalar (Section 5). The archived drive/brake data are a much harder case, and they are the ones our re-runs had to use. Restricting the archived TTC data to its 202 near-free-rolling samples, we fitted the nine active MF parameters by bounded least squares from 60 random starts ($C_y \in [1, 2]$, $E_y \leq 0.99$, other bounds wide). Results:

- The best fit (RMSE 55 N) puts p_{Cy1} *on its lower bound* 1.0. The data prefer a curve that is still rising at 6° and has no peak before 30°. With three slip angles, one of them zero, the peak slip angle is *not identifiable*.

- Among 34 near-optimal solutions (RSS within 5% of the best), p_{Ky1} ranges over [52.8, 135.7] and p_{Ky2} over [3.3, 9.2]. The cornering-stiffness *law* is not identified, only the secant stiffness at 3°.

- The profile of p_{Dy1} is shallow. Its approximate 95% range ($\Delta\text{RSS} < 3.84\hat{\sigma}^2$) is [2.05, 2.45], and it is conditional on the unidentified shape; this is a profile-likelihood statement in the sense of Raue et al. (2009).

- The column-normalised Jacobian has condition number 1.3×10^3, and its weakest direction couples p_{Cy1} and p_{Dy1} ($\sin(Cx) \approx Cx$ when there is no peak).

For these files, the PINN fills the gaps with the priors: the C_y clamp $[1.40, 1.60]$, the p_{Ky1} wall at 55, and the p_{Ey2} box. The fitted values sit on these constraints: $p_{Cy1} = 1.40$ in runs A, E, F and G, and $p_{Ky1} = 54.96$–54.97 in runs C, D, E and G. Figure 13 shows that all fitted curves agree inside the 0–6° data band and diverge outside it.

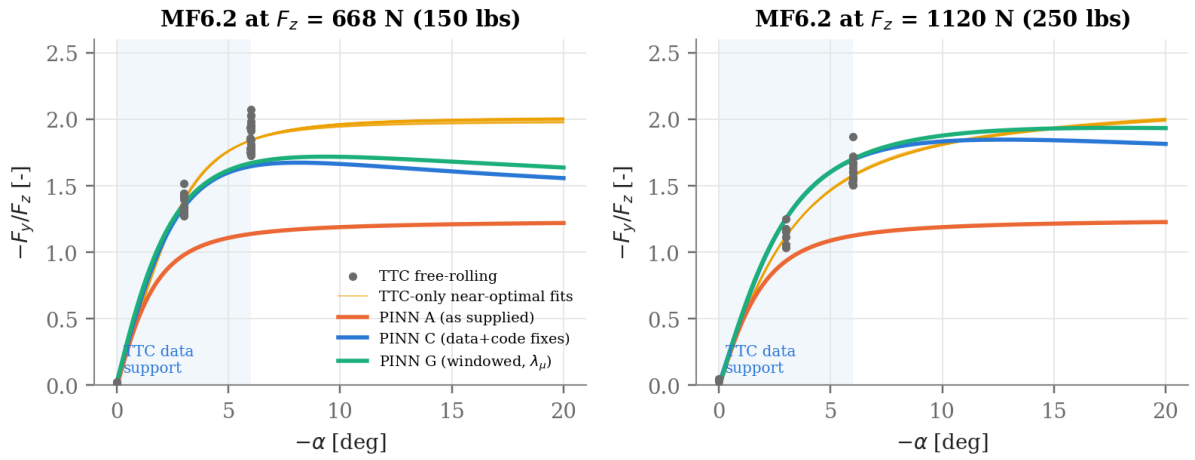


Figure 13: MF6.2 curves at two loads from the controlled re-runs on the archived TTC data. Grey dots: near-free-rolling samples. Thin yellow: 25 near-optimal TTC-only fits (spread = non-identifiability). Orange: run A, fitted to the combined-slip data. Blue/green: runs C and G.

6.4 Hessian spectrum and gradient conflict

The full-batch Hessian of $\mathcal{L}$ with respect to the 20 parameters (automatic differentiation, Figure 14b) has four eigenvalues that are exactly zero ($p_{Dy3}, p_{Hy3}, p_{Vy3}, p_{Vy4}$). The next one, 2×10^{-3}, belongs to p_{Ey4} and comes from the prior alone. The non-zero eigenvalues span up to eight orders of magnitude (run A). Run A also has one slightly negative eigenvalue (-7×10^{-5}), a saddle created by the clamps. The model is therefore over-parameterised by at least five parameters, and weakly determined in several more.

Figure 14a breaks the gradient at the run-C optimum down by loss term. For most parameters the terms *oppose each other*: the yaw balance pulls $v_{y,\mathrm{off}}$ up while the prior pulls it down, the lateral balance pushes p_{Dy2} positive against the degressive-friction prior, and the TTC and yaw terms disagree on p_{Ey1}, p_{Ky2} and p_{Cy1}. The optimum is a tug-of-war between data sources that are mutually inconsistent under the model, not a point where each is satisfied. This is the typical PINN loss-balancing pathology (Wang et al., 2021), and no choice of the weights $(w_{\mathrm{ttc}}, w_{\mathrm{lat}}, w_{\mathrm{yaw}})$ removes it. Weights move the compromise along the valley; they do not remove the valley.

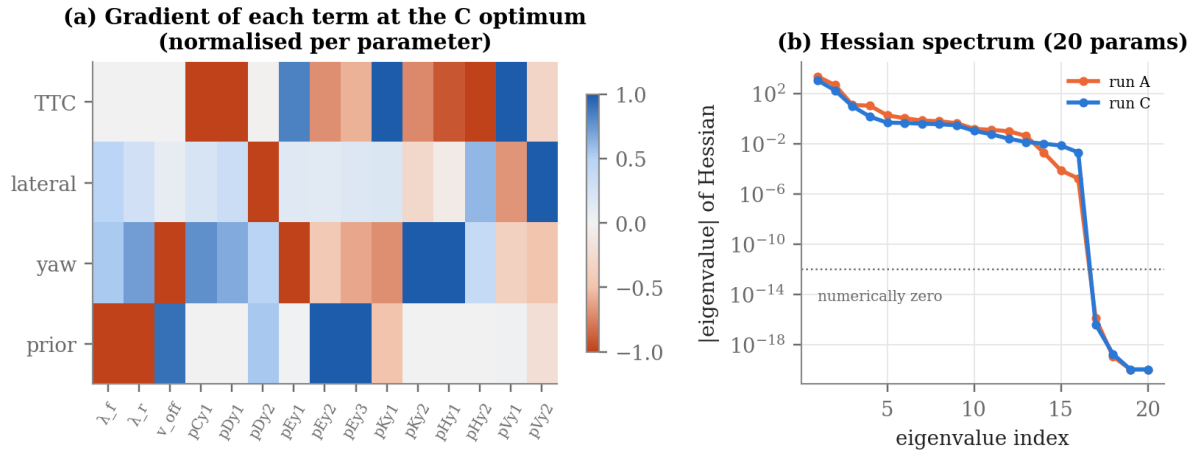


Figure 14: (a) Per-term gradient at the run-C optimum, normalised per parameter (camber parameters and p_{Ey4} omitted). Opposite colours in a column mean conflicting data sources. (b) Hessian eigenvalue magnitudes for runs A and C.

6.5 The steer-induced yaw moment has the wrong sign

A front-wheel force $F_{y,i}$, rotated by the steer angle δ, has a body-frame longitudinal component $-F_{y,i}\sin\delta$ at lateral position y_i. Its moment is $-y_i(-F_{y,i}\sin\delta)$. With FL at $y = -T_f/2$ (the placement used by both the slip kinematics (2) and the load-transfer model), this gives

$$M_\delta = \tfrac{1}{2}T_f\,(F_{y,FR} - F_{y,FL})\sin\delta, \tag{9}$$

whereas the code has $\frac{1}{2}T_f(F_{y,FL} - F_{y,FR})\sin\delta$, i.e. FL at $+T_f/2$. The inconsistency holds regardless of which global frame the data use, because it is *internal*: two blocks put FL on one side and the third puts it on the other. The term is not small. Its RMS is 38–51 N m, against 188–239 N m for the main axle moment and 144 N m for $I_z\dot{r}$, so the sign error injects about 80–100 N m of systematic yaw residual. Correcting it (run C versus B) changes $v_{y,\mathrm{off}}$ only from 1.51 to 1.49 m/s, which again shows that the degeneracy is driven by the data, not by this term.

7 Controlled Experiments

Each variant modifies the pipeline from the outside and changes one group of assumptions at a time. All variants use the code's default training configuration (Table 1) and the archived TTC data (the drive/brake set, Section 4). The TTC rows of Table 4 therefore describe those data. The vehicle-side conclusions do not depend on which rig data are used; the reference run shows the same offset and saturation with the cornering sweeps.

A Supplied code and data.

B Data fixes only: TTC free-rolling subset ($|F_X/F_Z| < 0.2$, 202 rows), 6 m runs removed, $|a_x| < 2$ m/s^2.

C B + code fixes: steer-moment sign, and keeping the L-BFGS end point.

D C + a learnable front steer scale (tests the steering-ratio hypothesis).

E C with $\lambda_{f,r}$ scaling *only the friction* D_y, so that $K_{y\alpha}$ is unscaled (MF-style λ_μ).

F E with $v_{y,\text{off}} \equiv 0$ (forces the kinematic slip angles).

G E on 2-s window means of the telemetry (81 windows; reduces high-frequency slip noise).

Table 4: Outcome of the variants (seed 42; seed 1 returns the same state for A and C).

	A	B	C	D	E	F	G
TTC rows / telemetry rows	7108/23238	202/11971	202/11971	202/11971	202/11971	202/11971	202/81†
TTC R^2	0.814	0.971	0.975	0.975	0.977	0.954	0.977
TTC RMSE [N]	211.6	105.8	98.3	98.4	94.2	132.4	93.6
$R^2(a_y)$	−0.008	0.212	0.233	0.234	0.267	−0.333	0.321
RMSE(a_y) [m/s^2]	2.99	2.25	2.22	2.22	2.17	2.92	1.64
RMS R_{yaw} (steady) [N m]	244	190	183	183	218	454	82
$v_{y,\text{off}}$ [m/s]	**2.502**	1.506	1.490	1.486	1.429	0 (fixed)	1.404
λ_f / λ_r	0.85/0.83	0.62/0.62	0.63/0.58	0.63/0.58	0.65/0.60	0.54/**0.50**	0.67/0.62
front steer scale	–	–	–	1.042	–	–	–
mean α_{FL} / α_{RL} [deg]	−13.8/ − 19.9	−11.3/ − 13.6	−11.2/ − 13.6	−11.4/ − 13.5	−10.9/ − 13.3	−3.5/ − 6.1	−11.0/ − 13.3
α_{peak} at F_{z0} [deg]	none	7.8	8.5	8.6	10.4	none	10.0
prior share of loss	28%	24%	25%	25%	25%	14%	35%

† window means, replicated ×40 so that batch statistics match the other runs. Bold: on a bound.

Reading the table.

- **Data fixes matter most** (A→B). Using the free-rolling TTC subset raises the TTC R^2 from 0.81 to 0.97 and gives the curve a peak again. Removing the mis-signed 6 m runs and the combined-slip samples lifts $R^2(a_y)$ from ≈ 0 to 0.21.
- **Code fixes are second order** (B→C): small, consistent improvements.
- **The steering ratio is not the missing degree of freedom** (D). Given the freedom, the optimiser changes the front steer by only 4% and keeps $v_{y,\text{off}}$.
- **Separating friction from stiffness helps** (E, G): better fits, $\lambda_\mu \approx 0.60$–0.67 (i.e. track $\mu \approx 1.2$–1.36 against the TTC ≈ 2.0). This is the physically expected parameterisation: the abrasive belt mostly inflates friction, while the carcass stiffness transfers.

- **Forcing kinematic slip angles fails** (F): λ_r hits its floor, E_y saturates and $R^2(a_y) = -0.33$. With the present V_y channel, the kinematic slip angles are not usable at face value.

- **Averaging does not remove the offset** (G). Its RMS yaw residual is the lowest (82 N m) and $R^2(a_y) = 0.32$ the highest, but $v_{y,\text{off}}$ stays at 1.4 m/s. The slip-angle error is dominated by low-frequency drift and run-to-run bias, which averaging cannot remove.

The persistence of $v_{y,\text{off}} \approx 1.4$–$1.5$ m/s in every variant is itself a result. It corresponds to 7° of sideslip at 11 m/s. Interpreted as a sensor lever arm, it would require $x_s \approx 1.4$ m, almost the wheelbase; interpreted as a heading misalignment, about 7°. Both are implausible as physical sensor errors, so $v_{y,\text{off}}$ should be read as a *nuisance parameter absorbing model and measurement error*, not as an identified sensor bias.

8 Roadmap

8.1 What the present data support

Defensible conclusions

1. **Rig tyre (TTC cornering sweeps):** well identified within ±12° and 217–1126 N. $R^2 = 0.994$, peak $\mu_y \approx 2.0$–2.7 with degressive load sensitivity, peak slip 7–11°, $K_{y\alpha} \approx 26$ kN/rad at 665.5 N. The peak level is uncertain to about ±0.15 in μ_y between trainings.
2. **Track grip level:** $\mu_{\text{axle}} \approx a_y/g = 0.92$–$1.18$ in steady state, i.e. a track-to-rig force ratio of 0.56–0.58 (whole-force scale) or 0.60–0.67 (friction-only scale). The on-track peak friction is $\mu \approx 1.2$–1.35.
3. **Not identified from telemetry:** the on-track operating slip angles, and therefore the on-track cornering stiffness, peak slip and understeer balance. The values implied by $v_{y,\text{off}} = 2.50$ m/s are artefacts of the saturation valley.
4. **Not identified at all:** the camber terms ($\gamma = 0$ everywhere). The asymmetry terms are identifiable from the cornering sweeps, but the priors hold them near zero.

8.2 Data

1. Keep the TTC cornering sweeps (5 loads, ±12°) versioned together with the code, and add the camber and pressure sweeps of the same round. Use the drive/brake runs, which are already archived, for a combined-slip MF6.2 (G_{yk}), the natural next step listed in the team's roadmap.

2. Measure lateral velocity with a dedicated sensor (optical ground-speed or dual-antenna GNSS), record its mounting position, and time-align all channels. Validate with the kinematic identity $a_y = \dot{V}_y + V_x r$ before any tyre fitting.

3. Fix the steering channel: sign convention, zero offset per session, and a measured steering-wheel-to-road-wheel map including Ackermann and compliance.

4. Add runs at several radii *and* speeds, so that the Ackermann angle and a_y are no longer collinear, and include slow ramp-steer runs that pass through the linear range.

8.3 Model and algorithm

1. **Errors-in-variables formulation.** Treat each quasi-steady segment's slip angles as latent variables α_k with a measurement model $\alpha_k^{\text{meas}} = \alpha_k + \varepsilon_k$, and estimate the tyre and the α_k jointly (total least squares, or a hierarchical Bayesian model). This removes the incentive to hide on the plateau.

2. **Parameterisation.** Estimate on track only what the track informs: λ_μ per axle (on D_y), possibly λ_K with a tight prior. Keep the TTC shape. Drop the camber, S_{Hy} and S_{Vy} parameters until both slip signs and camber sweeps are available.

3. **Smooth constraints.** Replace clamps with smooth maps into the admissible sets, so that the optimiser sees gradients everywhere and L-BFGS does not stop at kinks.

4. **Uncertainty.** Report the Laplace (inverse-Hessian) or MCMC posterior on the reduced parameter set. The Hessian machinery used in Section 6 is already in PyTorch.

5. **Validation protocol.** Hold out whole runs, not samples. Report R^2 of *run-mean* a_y and of the steady-state understeer gradient, the quantities the model will be used for.

9 Conclusion

The P26 PINN is a well-engineered piece of differentiable vehicle physics. The load transfer is faithful to the MATLAB model, the slip kinematics are consistent, the losses are thoughtfully normalised, float64 is used where it matters, and the Adam-then-L-BFGS schedule is sensible. On the rig data it does what it should: an excellent MF6.2 fit to the five TTC cornering sweeps ($R^2 = 0.994$), with degressive friction and peak slip angles of 7–11°.

The vehicle-level result is a different matter. The offset on its bound, the operating slip angles far beyond the peak and the constant predicted lateral acceleration are not

optimiser bugs. They are the least-squares answer to a question the telemetry cannot yet answer: the lateral velocity, and hence the slip angle, is measured less accurately than the effect it must resolve. Under that noise, the steady-state balances are best satisfied by putting every tyre on its plateau. Our controlled experiments confirm each link of this chain. They also show that fixing the 6 m steering sign, the yaw-moment sign, the grip parameterisation and the rig-data provenance improves every metric, but cannot on its own remove the degeneracy. What the current pipeline reliably delivers is the rig tyre and a track-to-rig grip ratio of about 0.56–0.67. With a validated lateral-velocity channel and an errors-in-variables formulation, the same code base is well placed to identify a fully track-calibrated MF6.2 tyre.

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A Identified Coefficients

Table 5: MF6.2 pure-lateral coefficients: initial values, the reference set (cornering sweeps, $F_{z0} = 665.5$ N) and the controlled re-runs A (defaults), C (data + code fixes) and G (windowed, friction-only scale), all on the archived drive/brake TTC data ($F_{z0} = 724$ N). Grey rows have zero data gradient.

Parameter	init	Reference	A	C	G	status (re-runs)
p_{Cy1}	1.50	1.6000	1.4000	1.4323	1.4000	clamp-bound (A, G)
p_{Dy1}	2.30	2.1505	1.4786	1.7040	2.0323	data (TTC $\approx$ 2.05–2.45)
p_{Dy2}	−0.12	−0.2705	0.0464	0.1991	0.1991	violates degressive prior
p_{Ey1}	−0.50	0.1911	1.0282	0.1613	0.3476	A: inadmissible ($E \to 0.99$)
p_{Ey2}	0	0.4670	0.1287	0.5096	0.5190	at +0.5 box
p_{Ey3}	0	−0.0121	0.0053	0.0112	0.0184	one-sided data
p_{Ky1}	70.0	65.86	64.20	54.97	54.96	C, G at 55 wall
p_{Ky2}	1.60	3.0118	3.2000	2.8428	2.3028	weakly identified
p_{Hy1}	0	−0.0037	−0.0059	−0.0009	−0.0074	
p_{Hy2}	0	−0.0028	−0.0018	0.0035	0.0032	
p_{Vy1}	0	0.0426	0.2246	0.0091	0.2872	large, sign-flipped on car
p_{Vy2}	0	0.0502	0.0264	−0.0782	−0.1480	
$p_{Dy3}, p_{Ey4}, p_{Hy3}, p_{Vy3}, p_{Vy4}$	0	0	0	0	0	structurally non-identifiable
λ_f / λ_r	1/1	0.561/0.576	0.852/0.826	0.633/0.583	0.669/0.625	(G: friction-only)
$v_{y,\text{off}}$ [m/s]	0	2.503	2.502	1.490	1.404	nuisance, not a sensor bias

B Output of the Default Re-run (Run A)

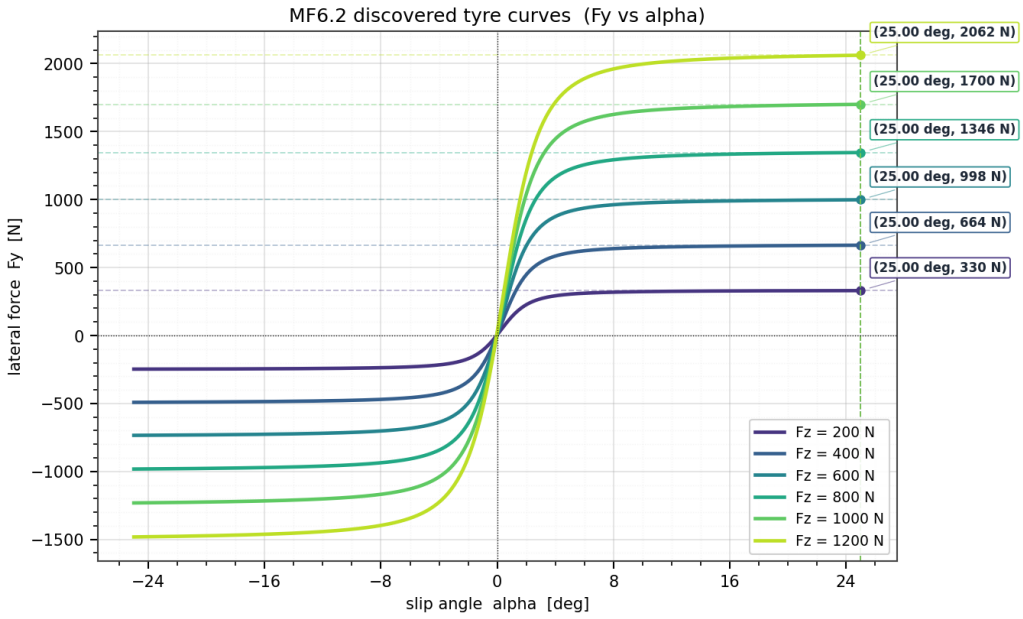


Figure 15: Tyre curves from re-run A (archived drive/brake TTC data): no peak within ±25° (every "peak" marker is at the grid end) and strong asymmetry from $p_{Vy1} = 0.22$.

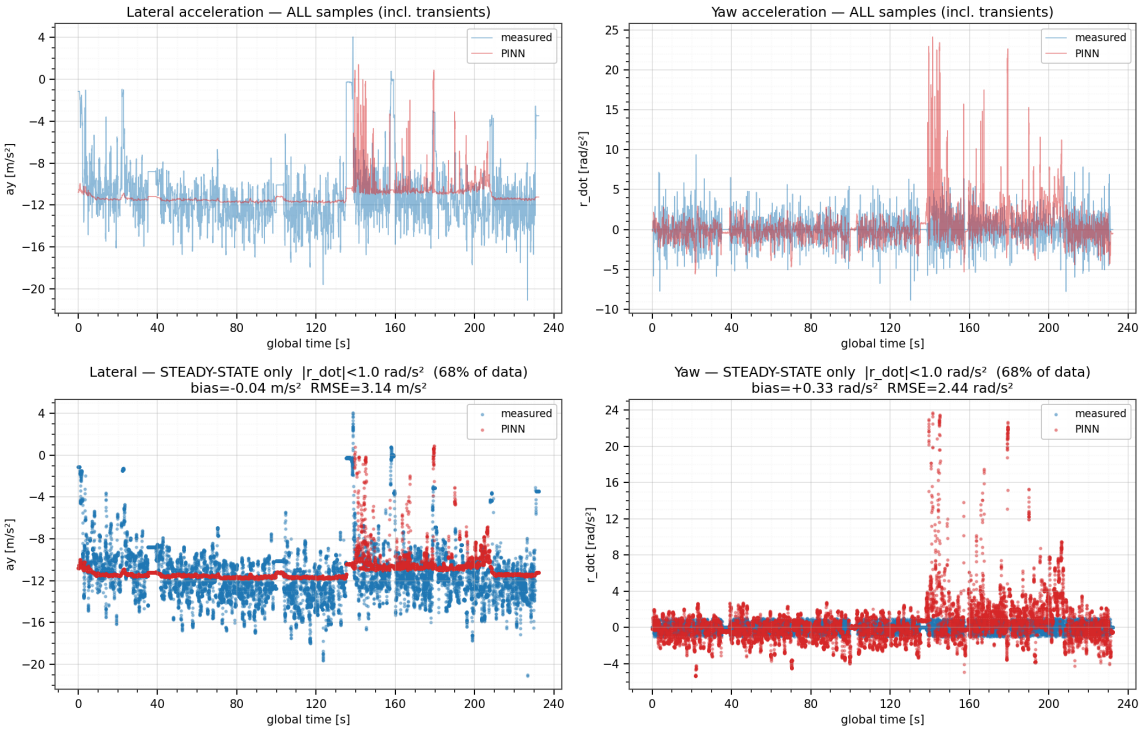


Figure 16: Vehicle balance from run A: the predicted a_y is almost constant (saturation). The yaw-acceleration spikes between 140 and 210 s are the 6 m runs with the mis-signed steering channel.

C Experiment Harness

All variants use the unmodified pipeline and replace individual functions from the outside, for example the corrected yaw residual:

```
def vehicle_residuals(self, Vx, Vy, r, r_dot, delta, ax, ay):
    pred = self.predict_wheels(Vx, Vy, r, delta, ax, ay)
    FL, FR, RL, RR = pred["Fy"]
    cd, sd = torch.cos(delta), torch.sin(delta); p = self.p
    Fy_body = (FL + FR) * cd + (RL + RR)
    # FL sits at y = -Tf/2 in slip_angles() and LoadTransfer
    Mz = p.lf*(FL + FR)*cd - p.lr*(RL + RR) + 0.5*p.Tf*(FR - FL)*sd
    return p.m_tot*ay - Fy_body, p.Iz*r_dot - Mz, pred
M.VehiclePINN.vehicle_residuals = vehicle_residuals
```

and the friction-only axle scale of runs E–G, which multiplies D_y inside the Magic Formula (so $B_y = K_{y\alpha}/(C_y D_y \lambda)$ increases and the cornering stiffness is preserved) instead of multiplying the output force.

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